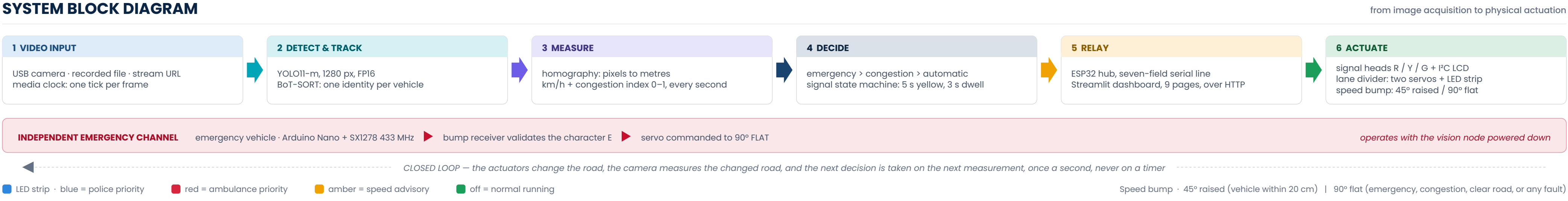


COMPUTER VISION-BASED SMART TRAFFIC MANAGEMENT

One camera measures the road in km/h — and the traffic signals, a movable lane divider and a retractable speed bump all act on that one measurement.

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SYSTEM BLOCK DIAGRAM



PROBLEM STATEMENT

Urban junctions are still governed by **fixed-time plans**. A timer holds a green over an empty approach and a red over a standing queue, because the junction has no measurement of the road it controls.

- **Nothing measures the road.** Timing, lane allocation and calming are all set open-loop, from a plan rather than from the traffic present.
- **Calming is indiscriminate.** A permanent speed bump penalises the one vehicle that must not be delayed — the ambulance behind it.
- **The parts share no measurement.** Signal, divider and bump are separate installations, each acting on its own assumptions.

In short: urban traffic control is driven by plans instead of measurements, because nothing on the road measures the traffic and shares that measurement with the devices that act on it.

AIM AND OBJECTIVES

Aim. To design, build and validate a camera-driven prototype that measures real traffic in engineering units, and uses that single measurement to drive adaptive signals, a movable lane divider and a retractable speed bump with emergency priority.

- 01 Report speed in km/h to within ±5% of ground truth.
- 02 Produce one congestion index on 0–1, updated every second.
- 03 Drive real signal hardware with no phase flicker.
- 04 Give priority only to a vehicle both liveried and flashing.
- 05 Raise the bump at 20 cm; flatten it on a validated radio command.
- 06 Move the divider, and show system state on LED strip and LCD.
- 07 Log every second; validate with 800+ automated checks.

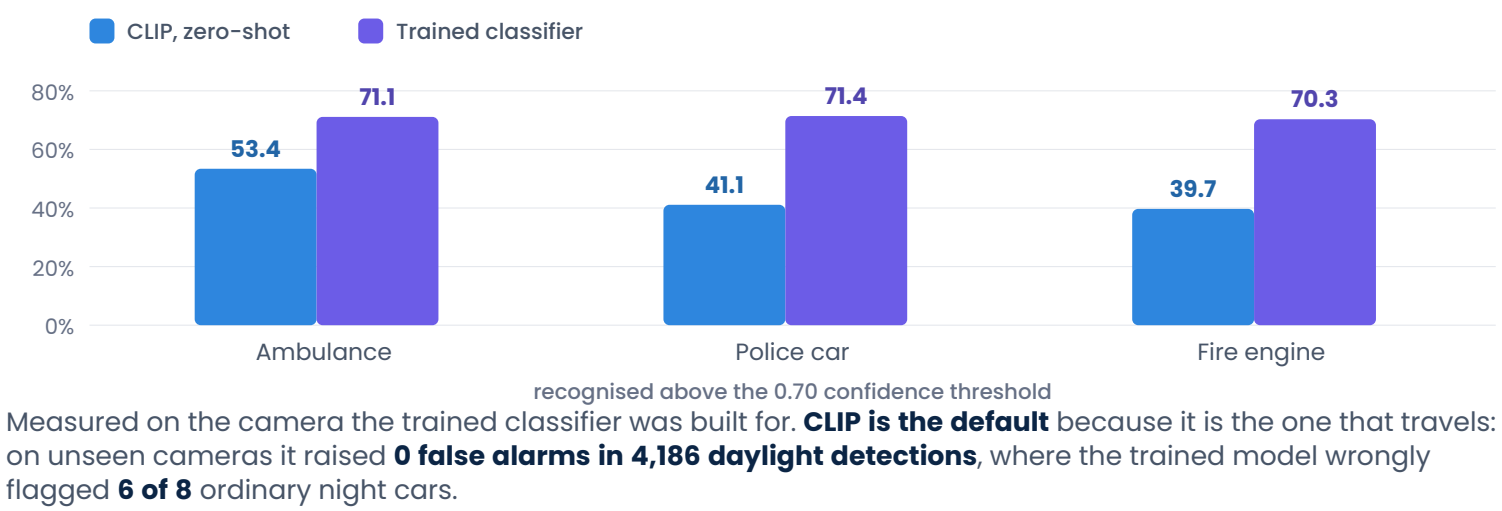
JUSTIFICATION FOR THIS PROJECT

1.19 million road-traffic deaths a year worldwide (WHO), and congestion costs cities working hours and fuel every day.

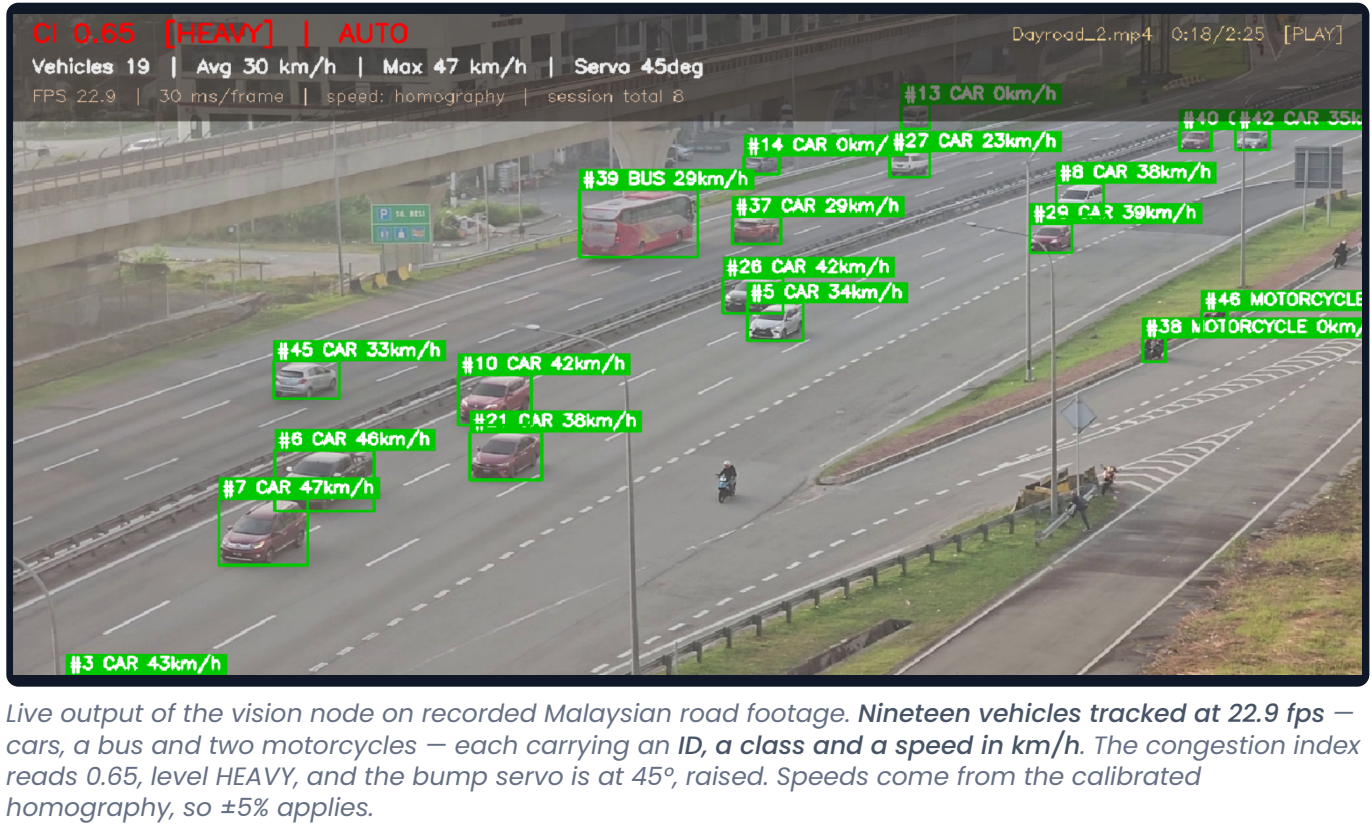
7–10% per minute is how fast survival falls after out-of-hospital cardiac arrest. A speed bump in front of an ambulance is not a small inconvenience.

RM 960 of parts, against excavation, a lane closure and a permanent installation for one inductive loop — reusing cameras many junctions already have.

AI MODEL BENCHMARKS



LIVE VISION OUTPUT



Live output of the vision node on recorded Malaysian road footage. **Nineteen vehicles tracked at 22.9 fps** — cars, a bus and two motorcycles — each carrying an ID, a class and a speed in km/h. The congestion index reads 0.65, level HEAVY, and the bump servo is at 45°, raised. Speeds come from the calibrated homography, so ±5% applies.

SUSTAINABILITY

- **No excavation.** It retrofits onto cameras and poles that already exist, avoiding the road-cutting and lane closure an inductive loop needs.
- **Less idling.** Signals that answer the queue, and a bump that flattens for an ambulance, both cut the stop-start running where urban fuel burn is worst.

3 GOOD HEALTH AND WELL-BEING

4 QUALITY EDUCATION

7 AFFORDABLE AND CLEAN ENERGY

8 DECENT WORK AND ECONOMIC GROWTH

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

10 REDUCED INEQUALITIES

11 SUSTAINABLE CITIES AND COMMUNITIES

12 RESPONSIBLE CONSUMPTION

13 CLIMATE ACTION

15 LIFE ON LAND

16 PEACE, JUSTICE AND STRONG INSTITUTIONS

17 PARTNERSHIPS FOR THE GOALS

PROPOSED METHODOLOGY

Six stages. No stage closed until its output could be **measured** — a servo angle against the physical plate, a speed against constructed ground truth, a command acknowledged on the serial monitor.

- 1 **Requirement analysis** — literature reviewed, measurable objectives fixed.
- 2 **System design** — vision, relay and actuation split; road and divider drawn in CAD; interfaces, pins and power allocated.
- 3 **Fabrication** — road printed, linkage 3D-printed, bump hinged to a servo; pipeline and three firmware images written.
- 4 **Subsystem testing** — servo travel, the 20 cm gate, the radio link, detection and signal timing each proved alone.
- 5 **Integration** — one measurement driving four actuators; priority ladder and power split settled here.
- 6 **Validation** — 833 automated checks, 69 recorded sessions, evaluation against 01–07.

THE THREE SUBSYSTEMS

- A VISION, SPEED & SIGNAL**
Displacement is measured on the road plane in metres, never in pixels. Two cues decide an emergency; CLIP asks **what** the vehicle is, the beacon detector asks whether it is **on a call right now**.
- B SMART RETRACTABLE SPEED BUMP**
An 8 × 4 cm hinged plate on a servo, calibrated against the plate itself: **90° is flat, 45° is raised**. Any fault or unknown state commands 90° — the position that cannot trap a vehicle.
- C ROAD PROTOTYPE & LANE DIVIDER**
A 360 × 480 mm printed carriageway with 30° merge lanes carries every device; two servos move the CAD-drawn, 3D-printed divider toward the busier direction.

INNOVATIVENESS, CREATIVITY & ENHANCEMENT

- **One measurement, four actuators.** The signal, the divider, the LED strip and the bump are not four gadgets — they are four consumers of a single congestion index and a single emergency flag.
- **Calming that gets out of the way.** A bump that flattens for a validated emergency command inverts the usual trade-off between calming and response time.
- **Two-cue emergency recognition.** Livery and beacon: a parked ambulance outside a hospital is still an ambulance, and it does not get priority.
- **A radio path that survives the computer.** The emergency link is peer-to-peer, so the safety-critical behaviour has the fewest dependencies of anything here.

TESTING & EVALUATION

UNIT	servo sweep, ultrasonic against a tape measure, LoRa register read-back
SUBSYSTEM	priority truth table, signal timing on a synthetic clock, detector benchmark
INTEGRATION	one measurement driving all four actuators, every command acknowledged
SYSTEM	69 recorded sessions, day and night footage, seven divider commands

SPEED, AGAINST CONSTRUCTED GROUND TRUTH		
TEST	EXPECTED	MEASURED
Homography mode	18 / 36 / 72	18.00 / 36.00 / 72.00
Parked vehicle	0 km/h	0.12
Clip at 1×, 2×, ½×	16.2 km/h	16.2

AGAINST THE OBJECTIVES

MET	01 18 / 36 / 72 km/h returned exactly; ±5% calibrated in the field
MET	02 12,919 one-second samples; index falls monotonically with speed
MET	03 5 s yellow and 3 s dwell exact against a synthetic clock
MET	04 night false alarms: 88 of 299 vehicles (29%) down to zero
MET	05 raised at 20 cm, flattened on a validated E; priority holds
MET	06 all seven divider commands answered, with LED and LCD state
MET	07 831 of 833 checks in one run; 21,526 vehicles archived
PARTLY	the dashboard's analog congestion hint is intermittent
PARTLY	radio shown at ~10 cm on the bench; no field-range test

MARKETABILITY & ACTUAL COST

	3 marks
Arduino Uno R3 × 3 — signal, lane and bump controllers	144.00
Arduino Nano + SX1278 433 MHz LoRa pair (TX + RX)	116.00
ESP32 DevKit V1 — relay hub	48.00
Servos × 3 — MG90S bump, 2 × SG90 divider	60.00
HC-SR04 ultrasonic, signal-head LEDs, resistors, button	62.00
WS2812B LED strip 1 m + 2 × 16×2 I²C LCD	106.00
3D-printed parts — divider linkage, servo mounts, bump hinge	158.00
Printed road mat 360 × 480 mm and bump plate	96.00
5 V 2 A supply, cabling, breadboard, fasteners, consumables	170.00
TOTAL COST INCURRED	RM 960.00

Malaysian retail prices, actual spend. Laptop, camera and workshop tools were existing assets and carry no project cost.

Market — city councils retrofitting junctions where CCTV already exists and a loop would mean digging; then hospital and fire-station corridors, campuses and gated estates.

Route — one instrumented junction as a paid pilot, a measured before-and-after report, then tender. The three subsystems sell separately.

ON FOOTAGE NEVER USED IN DEVELOPMENT

Over 21,083 frames and **2,511 tracked vehicles**, the recogniser singled out **166 (6.6%)**: 60 fire trucks, 58 ambulances, 47 police cars, mostly at 0.93–1.00 confidence. Only 21 reached beacon-confirmed priority; in daylight a beacon bright enough to register blooms to white, and white fails the red/blue purity gates by design.

SAFETY, PRACTICE & HONEST LIMITS

- **5 V SELV only.** No mains wiring anywhere in the prototype.
- **Actuators on their own supply.** Servo stall current is kept off the logic rail.
- **The bump fails flat.** Power loss, link loss and any unknown state all command 90°.
- **Ethical data.** Counts, speeds and classes only — no faces, no plates, no retained footage.
- **Measure, then claim.** Every figure here has a stated method, including what it does not prove — no mAP is quoted because no frames were hand-labelled.
- **833 checks across 7 suites.** The honest best single run is 831, and why the last two cannot pass together is documented rather than hidden.

Stated plainly: the dashboard's analog hint path is unreliable; the radio has only been shown across a bench; the plate and micro servo carry no vehicle load; and with no hand-labelled frames there is no mAP to quote. **Next:** a framed acknowledged link, a load-rated actuator, angle feedback, and a proper range test.